25it037

Practical 3

● AIM:

2. Duplicate card check (Card.java + Driver.java)

(a) Create class Card with private String rank, suit and a constructor.

(b) Override toString() to return rank + " of " + suit (e.g. "Queen of Hearts").

(c) Override equals()/hashCode() by rank + suit.

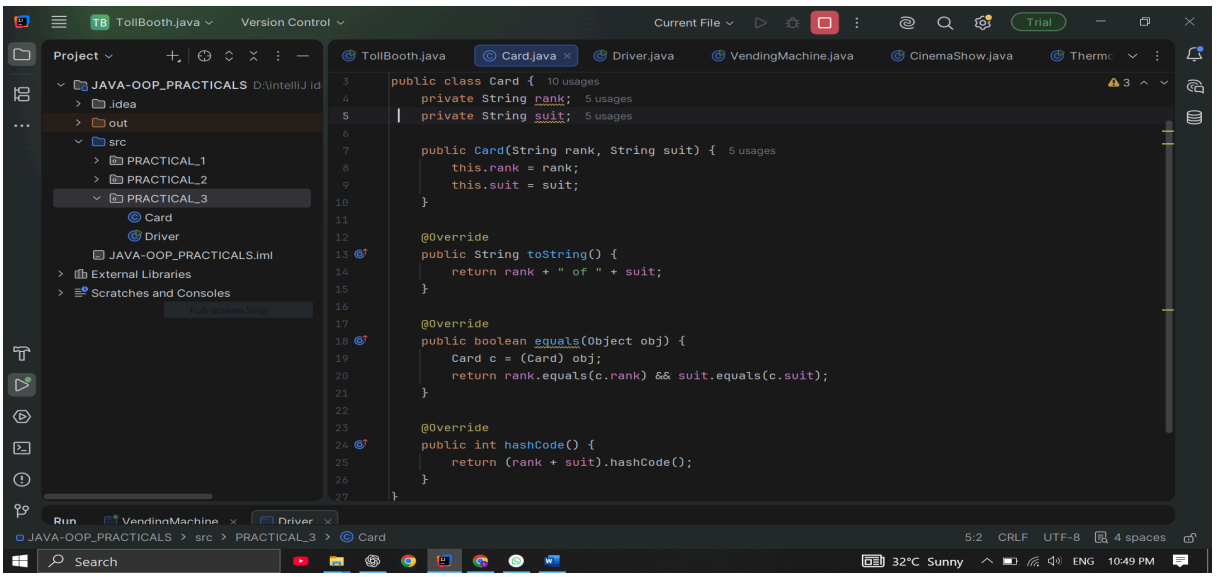
(d) In Driver.main: add cards to a Card[] one at a time; before adding each, use equals()

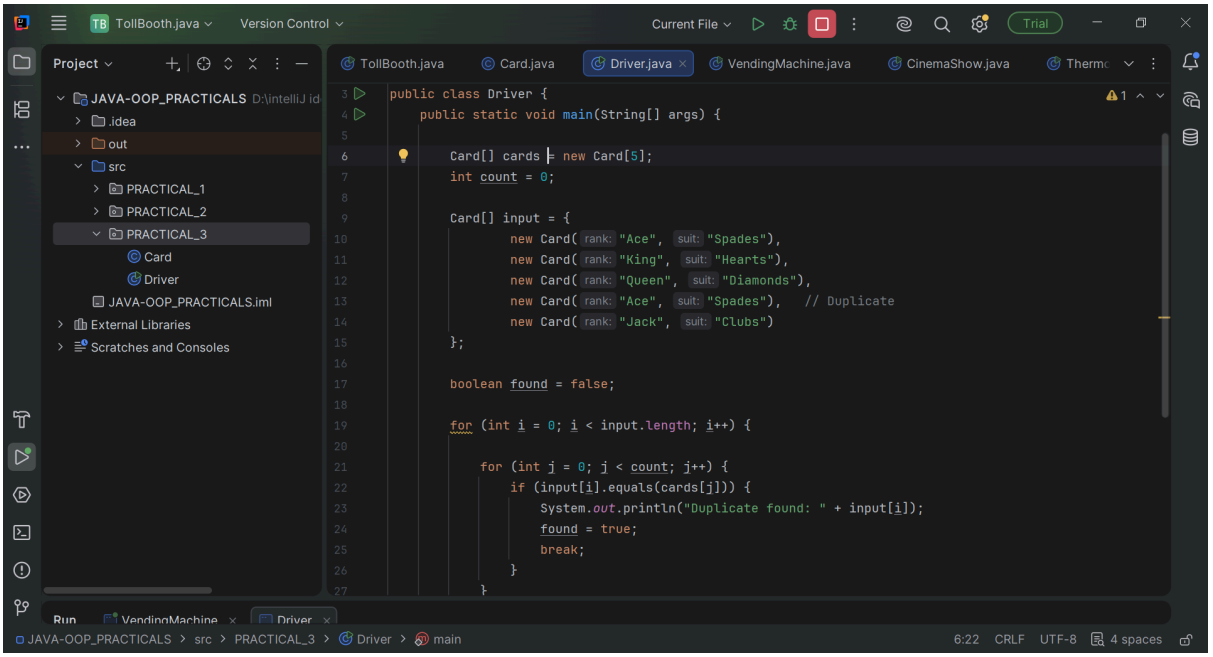
to check it against the earlier cards.

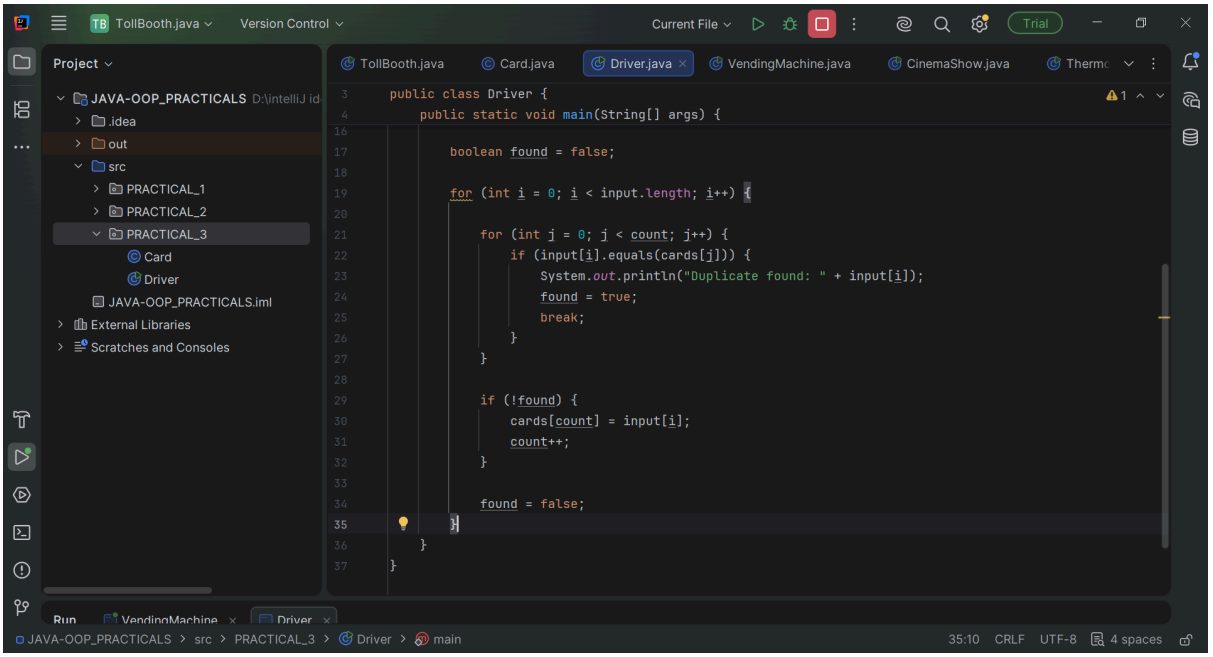
(e) Print "Duplicate found: <card>" the first time a repeat appears.

Expected output: a second "Ace of Spades" → "Duplicate found: Ace of Spades".

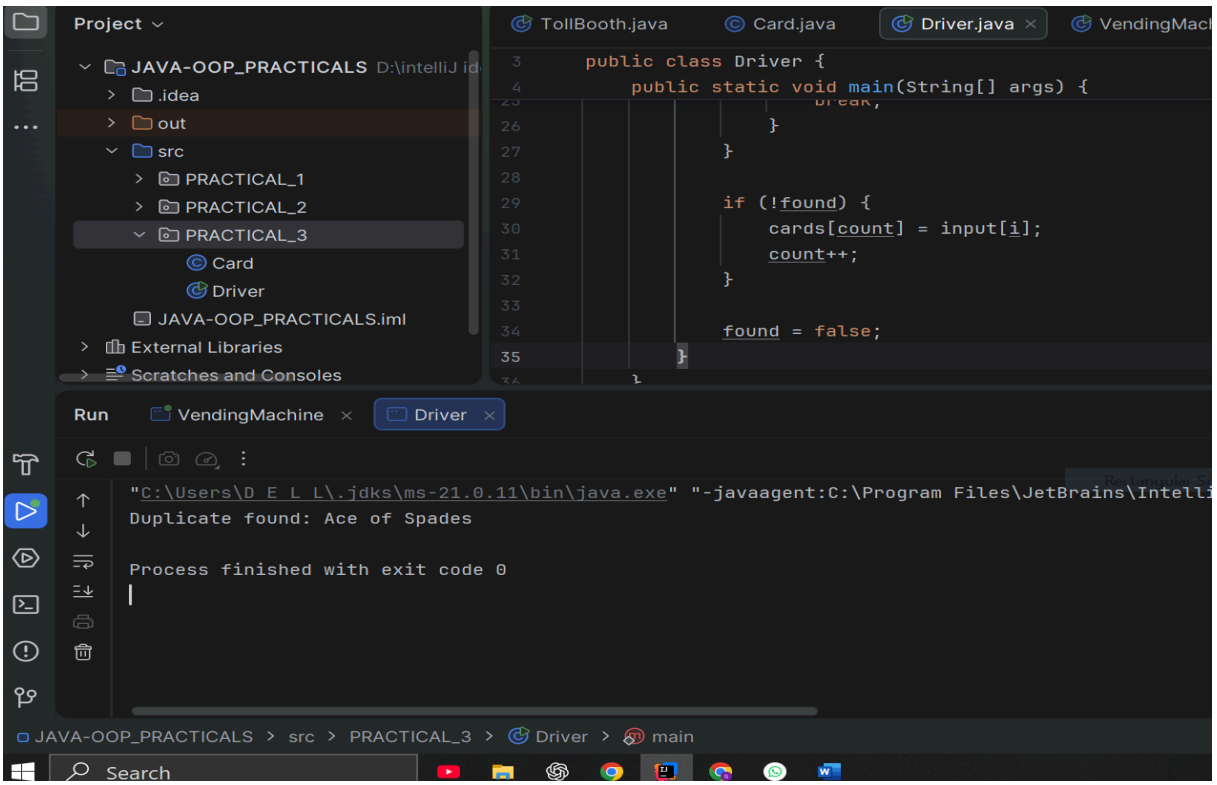
● Code Screenshot:


TollBooth.java
Version Control
Current File
Trial
Project
JAVA-OOP_PRACTICALS D:\IntelliJ id
.idea
out
src
PRACTICAL_1
PRACTICAL_2
PRACTICAL_3
Card
Driver
JAVA-OOP_PRACTICALS.iml
External Libraries
Scratches and Consoles
TollBooth.java
Card.java
Driver.java
VendingMachine.java
CinemaShow.java
public class Card { 10 usages
private String rank; 5 usages
private String suit; 5 usages
public Card(String rank, String suit) { 5 usages
this.rank = rank;
this.suit = suit;
@Override
public String toString() {
return rank + " of " + suit;
@Override
public boolean equals(Object obj) {
Card c = (Card) obj;
return rank.equals(c.rank) && suit.equals(c.suit);
@Override
public int hashCode() {
return (rank + suit).hashCode();
Run
VendingMachine
Driver
JAVA-OOP_PRACTICALS > src > PRACTICAL_3 > Card
5:2 CRLF UTF-8 4 spaces
Search
32°C Sunny
ENG 10:49 PM


TollBooth.java
Version Control
Current File
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JAVA-OOP_PRACTICALS D:\IntelliJ id
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PRACTICAL_1
PRACTICAL_2
PRACTICAL_3
Card
Driver
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External Libraries
Scratches and Consoles
TollBooth.java
Card.java
Driver.java
VendingMachine.java
CinemaShow.java
public class Driver {
public static void main(String[] args) {
Card[] cards = new Card[5];
int count = 0;
Card[] input = {
new Card(rank: "Ace", suit: "Spades"),
new Card(rank: "King", suit: "Hearts"),
new Card(rank: "Queen", suit: "Diamonds"),
new Card(rank: "Ace", suit: "Spades"), // Duplicate
new Card(rank: "Jack", suit: "Clubs")
};
boolean found = false;
for (int i = 0; i < input.length; i++) {
for (int j = 0; j < count; j++) {
if (input[i].equals(cards[j])) {
System.out.println("Duplicate found: " + input[i]);
found = true;
break;
Run
VendingMachine
Driver
JAVA-OOP_PRACTICALS > src > PRACTICAL_3 > Driver > main
6:22 CRLF UTF-8 4 spaces



- Output Screenshot:



● Learning Outcome:

By performing this practical, I learned how to create a user-defined class in Java with private data members and a constructor. I understood the importance of overriding the `toString()`, `equals()`, and `hashCode()` methods to represent objects meaningfully and compare them based on their content rather than their memory location. I also learned how to use arrays and loops to identify duplicate objects efficiently.

This is a good balance: detailed enough to earn marks, but concise enough for a practical file.